

1. Identification

Product Identifier: **RU-430 SP Liquid Rubber Part A**
 Product Code(s): SPRU430SPA
 Use: Component for Polyurethane Mold Rubber. For Industrial/Professional use only.
 Manufacturer: Polytek Development Corp.
 55 Hilton St., Easton, PA 18042 USA
 Phone Number: 800-858-5990 (8 a.m. to 6:30 p.m. EST)
 Emergency Phone: CHEMTREC 800-424-9300 or +1 703-527-3887
 E-mail: sds@polytek.com

2. Hazards Identification

GHS Classification as per 29 CFR 1910.1200:

GHS Classification:

Acute Toxicity (Inhalation) – Category 4
 Skin Irritation – Category 2
 Eye Irritation – Category 2A
 Respiratory Sensitization – Category 1
 Skin Sensitization – Category 1
 Carcinogenicity – Category 2

Label Elements: Danger



Hazard Phrases

H315 Causes skin irritation.
 H317 May cause an allergic skin reaction.
 H319 Causes serious eye irritation.
 H332 Harmful if inhaled.
 H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.
 H351 Suspected of causing cancer.

Precautionary Phrases

P203 Obtain, read and follow all safety instructions before use.
 P261 Avoid breathing fumes, vapors, mists or sprays.
 P264 Wash thoroughly after handling.
 P271 Use only outdoors or in a well-ventilated area.
 P272 Contaminated clothing should not be allowed out of the workplace.
 P280 Wear protective gloves, protective clothing, eye protection, and face protection.
 P284 In case of inadequate ventilation wear respiratory protection.
 P302+P352 IF ON SKIN: Wash with plenty of soap and water.
 P304+P340 IF INHALED: If breathing is difficult, remove person to fresh air and keep comfortable for breathing.
 P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
 P317 Get medical help.
 P318 IF exposed or concerned, get medical advice.
 P333+P317 If skin irritation or rash occurs: Get medical help.
 P337+P317 If eye irritation persists: Get medical help.
 P342+P316 If experiencing respiratory symptoms: Get emergency medical help immediately.
 P362+P364 Take off contaminated clothing and wash before reuse.
 P405 Store locked up.
 P501 Dispose of contents and container in accordance with local, regional and national regulations.

Supplemental Information: Individuals sensitized to isocyanates should discontinue use. Long-term overexposure to isocyanates may cause lung damage. This is one part of a two-part system. Read and understand the hazard information on part B before using.

3. Composition/Information on Ingredients

Chemical Name	CAS #	%
Toluene Diisocyanate (TDI)	26471-62-5	≤2
Polyether polyol-TDI prepolymer	9057-91-4	85-90
Exact concentrations are withheld as trade secret. Other ingredients are not listed because they are either not hazardous or are below concentration/cut-off thresholds.		

4. First-Aid Measures

Eye Contact: Rinse thoroughly with water for at least 15 minutes, holding the eyelids open to be sure the material is washed out. Get medical attention if irritation develops or persists.

Skin Contact: Remove contaminated clothing. Wash contact area thoroughly with soap and water. Get medical attention if irritation develop or persist. Launder clothing before reuse. Discard items that cannot be decontaminated.

Inhalation: Remove person to fresh air. Give artificial respiration if needed. If breathing is difficult, oxygen should be administered by qualified personnel. Get immediate medical attention.

Ingestion: Do not induce vomiting unless directed to do so by medical personnel. Get medical attention if you feel unwell.

Most Important Symptoms/Effects: Causes skin and eye irritation. Vapors or mists may cause respiratory irritation. May cause allergic skin and/or respiratory reaction in sensitized persons. Symptoms include skin rash, wheezing, shortness of breath and other asthma symptoms.

Indication of Immediate Medical Attention/Special Treatment:

Immediate medical attention is required for asthmatic symptoms or serious inhalation exposures. Respiratory symptoms, including pulmonary edema, may be delayed. Persons receiving significant exposure should be observed 24-48 hours for signs of respiratory distress. Persons sensitized to Diisocyanates should consult a physician before working with respiratory irritants or sensitizers.

5. Fire-Fighting Measures

Extinguishing Media: Use water fog, foam, carbon dioxide or dry chemical. Do not use solid water stream. Solid stream of water into hot product may cause violent steam generation or eruption.

Specific Hazards: Not classified as flammable or combustible. Product will burn under fire conditions.

Special Protective Equipment & Precautions for Fire-Fighters: Wear positive pressure, self-contained breathing apparatus and full-body protective clothing. Cool fire-exposed containers with water.

6. Accidental Release Measures

Personal Precautions, Protective Equipment and Emergency

Procedures: Remove all ignition sources. Clear non-emergency personnel from the area. Ventilate area. Wear appropriate protective clothing to prevent eye and skin contact and respiratory protection.

Methods and Materials for Containment and Cleanup: Cover with an inert absorbent material and collect into an appropriate container for disposal. Do not seal the container since CO₂ is generated on contact with moisture and dangerous pressure buildup can occur. Decontaminate floor area with a mixture of water plus isopropyl alcohol (10-20%), household ammonia (10%), and detergent (2%).

7. Handling and Storage

Safe Handling: Avoid breathing fumes, vapors, mists or sprays. Use with adequate ventilation. Avoid contact with the eyes, skin and clothing. Wash thoroughly after handling. Do not eat, drink or smoke in the work area. Keep container closed when not in use.

Safe Storage: Store indoors at temperatures between 55°F and 95°F (13°C and 35°C). Store in original, unopened containers. Protect from

atmospheric moisture and water since TDI reacts with water to form CO₂ leading to potentially dangerous pressure build up in sealed containers.

8. Exposure Controls/Personal Protection

Occupational Exposure Limits: For TDI:

OSHA PEL 0.02 ppm (C)

ACGIH TLV 0.005 ppm TWA; 0.02 ppm STEL

Ventilation: Use with adequate general or local exhaust ventilation to maintain exposure levels below the occupational exposure limits.

Respiratory Protection: If needed (i.e., ventilation is inadequate), use a NIOSH-approved air-purifying, tight-fitting, half-face respirator with organic vapor cartridges. Respirator selection and use should be based on contaminant type, form and concentration. For higher exposures or in an emergency, use a supplied-air respirator. Use respirators in accordance with OSHA's Respiratory Protection Standard (29 CFR 1910.134).

Skin Protection: Wear impervious gloves, such as butyl rubber or nitrile rubber.

Eye Protection: Wear chemical safety goggles/glasses.

Other Protective Measures: Wear impervious clothing to prevent skin contact and contamination of personal clothing. An eye wash and washing facility should be available in the work area. Follow good Industrial Hygiene practices.

9. Physical and Chemical Properties

Appearance: Clear pale yellow to amber liquid

Odor: Pungent, slightly sweet

Odor Threshold: Not determined

pH: Not applicable

Melting Point: No data available

Boiling Point: No data available

Flash Point: >350°F (177°C) estimated

Evap. Rate: No data available

Flammable Limits: No data available

Vapor Pressure: ≤0.01 mm Hg @ 20°C

Vapor Density: No data available

Relative Density: 1.05 @ 25°C

Solubility: Insoluble in water

Partition Coefficient: n-octanol/Water: Reacts with water

Auto-Ignition Temp: No data available

Decomposition Temp: No data available

Viscosity: 5,000-10,000 cP

10. Stability and Reactivity

Reactivity: Diisocyanates react with many materials and the rate of reaction increases with temperature. Reaction with water generates carbon dioxide and heat.

Chemical Stability: Stable under recommended conditions.

Possibility of Hazardous Reactions: Elevated temperatures can cause hazardous polymerization. Polymerization can be catalyzed by strong bases or water. Reaction with water generates carbon dioxide, and results in heat and pressure buildup in closed systems.

Conditions to Avoid: Avoid moisture and temperatures below 55°F (13°C) and above 95°F (35°C) to protect product integrity.

Incompatible Materials: Avoid contact with water, acids, bases, alcohols, strong oxidizers, and some metals (e.g., aluminum, zinc, brass, tin, copper).

Hazardous Decomposition Products: Possibly isocyanate vapor, carbon monoxide, nitrogen oxides, and traces of hydrogen cyanide.

11. Toxicological Information

Eye Contact: Causes serious eye irritation.

Skin Contact: Causes skin irritation. May cause an allergic skin reaction.

Inhalation: At room temperature, vapors are minimal due to low volatility. Vapors or aerosols (e.g., generated during heating or spraying) may cause respiratory irritation and possibly pulmonary edema, or respiratory sensitization. For individuals sensitized to TDI, exposure may result in allergic respiratory reactions (e.g., coughing, wheezing, difficulty breathing).

Ingestion: No data available.

Chronic Health Effects: Repeated or prolonged exposure to isocyanates may cause an allergic sensitization of the respiratory tract causing an asthma-like response upon re-exposure. Repeated overexposure to isocyanates has been associated with decreased lung function. Repeated or prolonged dermal contact with this product may cause allergic skin sensitization in some individuals. No test data. Product is not expected to be a mutagen or reproductive toxin.

Acute Toxicity Values: For TDI:

Oral LD₅₀ (rat): >2,000 mg/kg

Dermal LD₅₀ (rabbit): >9,400 mg/kg

Inhalation LC₅₀ (rat): 0.24 mg/L/4 hr

Carcinogenicity: Suspected of causing cancer.

Specific Target Organ Toxicity: No data available.

12. Ecological Information

Poly 74A and 75A react with water to form insoluble polyureas.

Movement in the aquatic and terrestrial environment would be limited.

Product is not readily biodegradable and not expected to bioaccumulate.

13. Disposal Considerations

Dispose according to local, state and federal regulations. Upon exposure to moisture, product forms an inert, non-hazardous solid. In the U.S., this product is not a RCRA hazardous waste (per 40 CFR 261).

14. Transport Information

Not regulated for transport in any mode.

EMERGENCY SHIPPING: CHEMTREC, 800-424-9300 or +1-703-527-3887

15. Regulatory Information

U.S. FEDERAL REGULATIONS:

CERCLA Reportable Quantity: RQ for TDI is 100 lb. Some States have more stringent requirements. Report spills in accordance with local and state regulations.

SARA TITLE III Section 311/312: Acute Health, Chronic Health

Section 313 Toxic Chemicals: This product contains the following chemicals subject to SARA Title III Section 313 Reporting requirements:

Toluene Diisocyanate CAS 26471-62-5 ≤2%

EPA Toxic Substances Control Act (TSCA) Status: Components of this product are listed on the TSCA inventory.

STATE REGULATIONS:

California Proposition 65: WARNING: This product can expose you to chemicals including Toluene diisocyanate (TDI), which is known to the State of California to cause cancer and/or reproductive harm.

www.P65Warnings.ca.gov

16. Other Information

Training Advice: Train personnel using this product in proper chemical handling, engineering controls and protective equipment.

Recommended Uses and Restrictions: This product is intended for industrial/professional use only.

SDS Revision Notes: GHS Format: October 7, 2024

Disclaimer: The information contained herein is considered accurate; however, Polytek® Development Corp. makes no warranty regarding the accuracy of the information. The user must determine the suitability of the product for the intended use and accepts all risk and liability associated with that use.